

Research Reports on Computer Science

Smoke Test
Report

July 30, 2026 - 22:15
PM

CitationEdge Analysis Report

📘 Executive Summary

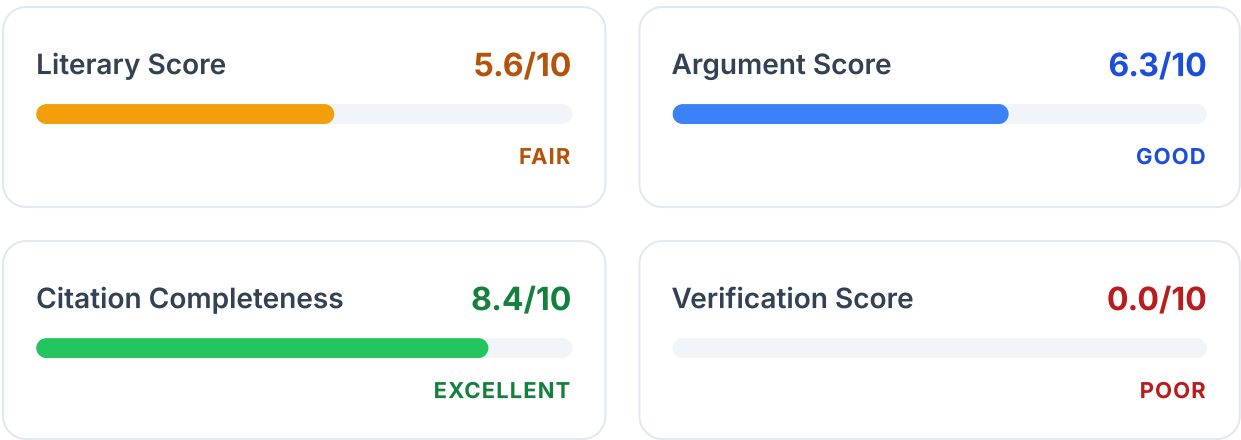
This executive summary provides an overview of a research paper analysis report focusing on task scheduling algorithms in cloud computing. The paper claims that IWDC has a better makespan and provides a comparison of task scheduling algorithms, both with high confidence levels. The counterfactuality analysis reveals a low overall score, indicating that the paper is grounded in empirical evidence and established scientific knowledge. However, there are concerns regarding the lack of specific details in one claim. The paper's strengths include high accuracy scores for both claims and a straightforward description of a common research topic. The citation gaps analysis suggests that the paper could benefit from exploring deep reinforcement learning, fog computing, and distributed control systems for optimizing resource allocation. The overall score of the paper is 5.08/10, with literary and argument scores being relatively low.

OVERALL

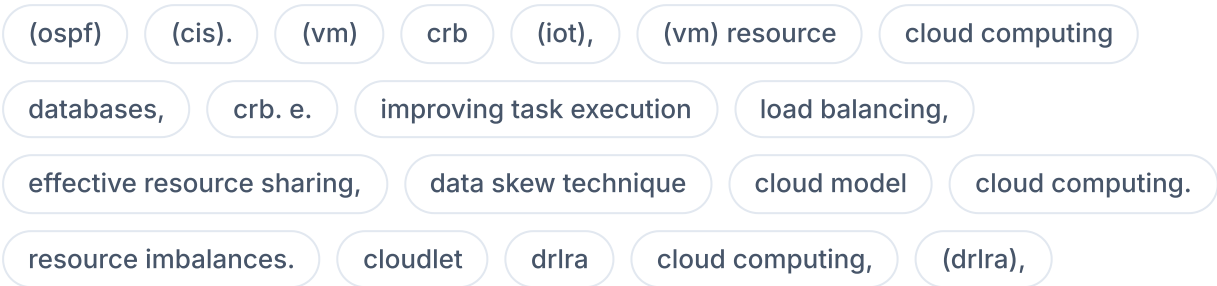
5.1/10

Fair

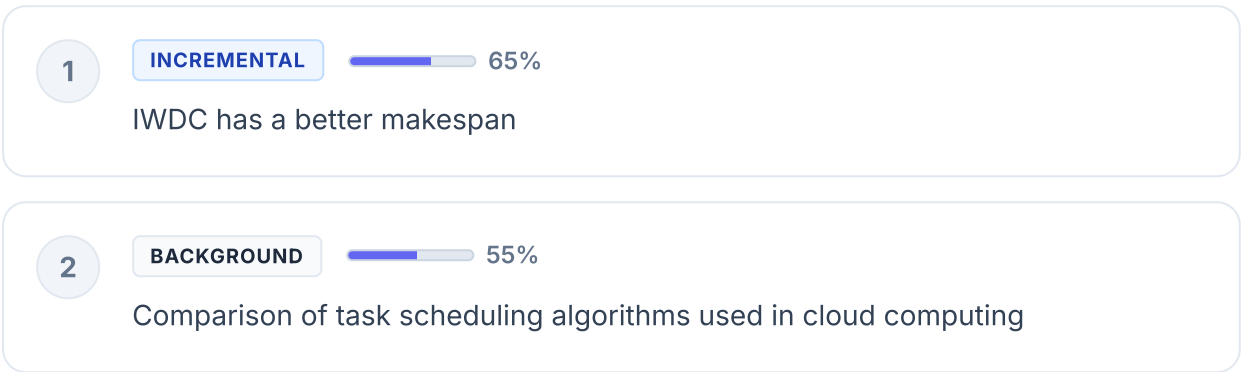
Quality Scores



Extracted Keywords



Research Claims



Claim Verification



#	Claim	Verdict	Reasoning
1	IWDC has a better makespan	UNVERIFIED	No relevant evidence found in external sources.
2	Comparison of task scheduling algorithms used in cloud computing	UNVERIFIED	No relevant evidence found in external sources.

Counterfactuality Analysis

0.10 ACCURACY SCORE		MOST COMMON TYPE FACTUAL	STRENGTHS ✓ one claim is a straightforward description of a common research topic ✓ high accuracy scores for both claims
Claim Analyzed	Type	Score	
IWDC has a better makespan	FACTUAL	0.20	
Comparison of task scheduling algorithms used in cloud computing	FACTUAL	0.00	

AI Text Detection

Likely Human-Written Verdict: Verdict.REAL		AI PROBABILITY 1.0%
Human		AI
Confidence: 98.0%	Method: [text_fastdetectgpt] Human-authored: 99.9% confidence	Processed in: 9696ms

Full Paper Text with AI Analysis

Each paragraph tagged with its AI-generation probability. Red = likely AI, Green = likely human.

Human: 0%

238 words

: Companies across industries increasingly depend upon cloud computing to manage their Industrial Internet of Things (IIoT) technology. Machines are connected over a network in the IIoT. Cloud computing plays an essential role by connecting people, devices, work processes, and buildings to deliver cloud services in industries. But cloud computing faces a problem with task scheduling, high latency delay, and memory management, affecting the overall cost of industries using cloud services. A major concern in the cloud computing field is task scheduling which is essential for achieving cost-effective execution and improving resource usage. It refers to assigning available resources to user tasks. This problem can be solved effectively by improving task execution and increasing the use of resources. The waiting time between a client's sent request and a cloud service provider to give a response, known as latency, is another issue in cloud environments. In cloud computing, this delay can be significantly higher. As a result, users of various cloud services may incur increased expenses due to this delay. Finally, among the most significant topics in cloud computing is efficient memory management, which handles integrated data and optimizes memory management algorithms. This paper proposes a cloud model for IIoT, which provides task scheduling, helps reduce latency, and optimizes memory management. This proposed model helps to reduce the cost of using cloud computing in IIoT. Keywords: task scheduling, cloud model, cost-effective, IIoT, Industry 5.0, effective resource sharing, energy optimization

Cloud computing is an internet-based computing service that provides services, like infrastructure, data storage, and applications remotely. Various computing resources, like storage, processing power, databases, networking, analytics, artificial intelligence, and software applications, are available through cloud computing. Companies do not need to purchase and maintain a physical, on-site Information technology (IT) infrastructure on their premises because they may outsource these resources and use computing assets whenever required. Customers can access new or scale existing resources as required by paying charges based on the number of resources used. Users pay for their services the same way as home utilities. Cloud computing resources are divided dynamically and assigned on demand. As illustrated in Figure 1, a cloud service provider offers computing resources to users. These computing resources are available on demand and are divided and assigned dynamically. Copyright ©2023 Satyajeet Shinge, et al. DOI:

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<https://creativecommons.org/licenses/by/4.0/> Special Issue|2023| 55 Research Reports on Computer Science User 1 Cloud provider R Internet Resource scheduler R R User 2 User 3
Figure 1. Resource allocation in cloud computing The Industrial Internet of Things (IIoT) refers to networked sensors, instruments, and other resources used in manufacturing and energy management that are integrated with industrial computer applications (Figure 2). This connectedness makes the ability to collect, trade, and analyze data possible, which may also positively affect the economy. The IIoT is an expansion of a distributed computing system that enhances and optimizes process controls using cloud computing and allows a higher level of automation. Sensor Automation Safety Industry 5.0 Monitoring Quality tool Indust

Human: 31%

304 words

for the cost-effective cloud model in IIoT. A series of researches are done in cloud computing on task scheduling, latency, and memory allocation to obtain cost-effective cloud computing. This paper proposes a cloud model for IIoT which combines a solution of task scheduling, latency, and memory allocation to minimize the computational cost of the industry. Figure 4 shows the proposed system for an optimized cloud model to achieve better performance by adopting the task scheduling algorithm, memory management techniques, and cloud-edge nodes to reduce latency. The adaptive model employs the data skew technique on IIoT application data and employs the cloud model to allocate the resource by migrating the resources based on the load. The proposed method organizes the optimization technique to achieve load balancing; this proposed cloud model employs an optimization technique to minimize the load and improves the computation efficiency. Various software and hardware components comprise a computational cloud environment, including computing nodes, storage systems, databases, network resources, file systems, etc. It is made up of four main parts: the cloud user, the cloud resources, the cloud resource broker (CRB), and the cloud information service (CIS). When using the cloud, a user first communicates with a resource broker to transmit a task for computation. Following that, resource finding, Research Reports on Computer Science 58 | Satyajeet Shinge, et al. scheduling techniques, and task processing are carried out. Finally, the CIS was employed as an agent. It gathers all pertinent data, like resource availability, node capacity, etc., and gives it to the resource breaker so that they may decide when to schedule things. The interactions between various clouds components are done using multiple steps which are as follows: a. The cloud user runs their application and after analysis and specifying their requirement, submits their jobs to CRB. b. The CRB collects all resour

Human: 4%

95 words

and discussion Table 4 gives the comparative analysis of all task scheduling algorithms. These algorithms are compared based on makespan value, resource utilization, response time, energy consumption, and throughput. Effective task scheduling algorithms should have less makespan, balanced resource utilization, less response time, low energy consumption, and more throughput value. The comparison shows that the IWDC algorithm gives a better makespan value as compared to other algorithms. It also gives good resource utilization, minimum response time, and large throughput. So, this algorithm can be used in the presented model to develop a cost-effective cloud model.

AI: 96%

297 words

This paper presents a cost-effective cloud model for IIoT. This proposed model can help to minimize industrial computational cost by providing efficient task management, reducing latency delay, and optimizing memory allocation. This paper identifies various work scheduling techniques applied in cloud computing and compares them all. Results and discussion show that each algorithm has distinct advantages for parameters like makespan, response time, resource utilization, energy consumption, and throughput. But the IWDC algorithm gives good results for each parameter. If the IWDC algorithm is used for task scheduling, then it helps to reduce the cost of computing in IIoT. If the cost of computation in industries is reduced, it benefits society by reducing the cost of products or services. Society can get services and products from industry at minimum cost. This paper also presents the evaluation done in Industries till the current year. Manufacturers are now including new technologies into their manufacturing facilities and operations, like the IoT, robotics, artificial intelligence, cloud computing, and machine learning to enable fast and efficient production of goods and services. The presented model will help towards Industry 5.0. Special Issue|2023| 61 Research Reports on Computer Science Research Reports on Computer Science 62 | Satyajeet Shinge, et al. high. excellent throughput. is more than throughput is are more in numbers. is faster. The result is fast execution of tasks. having less execution time with more execution time executed is more. algorithms, this algorithm Throughput This algorithm provides It gives good throughput. If the number of tasks Throughput is low if tasks The number of tasks Compared to other energy. other algorithms. consumption. switches, it consumes more heterogeneous tasks. heterogeneous tasks. efficiency as compared to more energy. Energy consumption FCFS has low energy Due to more context Consumes more energy for Consumes more energy

Uncertain: 47%

43 words

in better makespan. It gives the minimum IWDC has a better Parameters FCFS RR Min-Min Max-Min PSO IWDC Algorithm Table 4. Comparison of task scheduling algorithms used in cloud computing Conflict of interest The authors declare that there is no conflict of interest.

Citation Gaps



MEDIUM SEVERITY

There is a need to explore the application of deep reinforcement learning for resource allocation in mobile edge computing

**MEDIUM SEVERITY**

The paper could benefit from discussing priority-aware dynamic resource allocation for IoT networks using edge-based approaches

**MEDIUM SEVERITY**

The paper does not examine the potential of fog computing and cloudlet-based architectures for optimizing resource allocation in IoT and IIoT scenarios

**LOW SEVERITY**

The integration of deep reinforcement learning and federated learning for intelligent resource allocation in mobile edge computing is not explored

**LOW SEVERITY**

There is a lack of discussion on the impact of data skew technique on resource allocation in cloud computing and its potential mitigation using DRLRA

**LOW SEVERITY**

The paper could be strengthened by exploring the application of distributed control systems and ethernet-based networks for optimizing resource allocation in cloud deployments

**HIGH SEVERITY**

The paper lacks a comprehensive review of existing deep reinforcement learning architectures for resource allocation in cloud computing and IoT networks

Argumentation Analysis

The argumentation structure is moderately strong, with a clear comparison of task scheduling algorithms and a confident claim about IWDC's makespan. However, the lack of counterargument handling and limited novelty in the background claim are notable weaknesses.



Strengths

- . Clear comparison of task scheduling algorithms
- . Confident claim about IWDC's makespan



Weaknesses

- . Lack of counterargument handling
- . Limited novelty in the background claim